Review of “QAGT-MLP: AN ATTENTION-BASED GRAPH TRANSFORMER FOR SMALL AND LARGE-SCALE QUANTUM ERROR MITIGATION”

The authors present a machine-learning based approach to error mitigation, implementing a graph transformer which combines global and local attention, whose output is used together with circuit features to predict the noise-mitigated values.
The authors test their model on the dataset from H. Liao *et al*., *Nature Machine Intelligence* 6, 1478-1486 (2024), comparing with the best model of that work, reporting an improvement in the predicted values.

Overall, the machine learning architecture and the approach themselves are not new, but they represent more a marginal improvement of the graph neural architecture proposed in H. Liao *et al*., *Nature Machine Intelligence* 6, 1478-1486 (2024) with the addition of the attention mechanism.

The paper is often unclear, the results are minimal and the work itself does not represent a notable advance in the field to justify publication in *Nature communications ai & computing*.
I shall also mention that the Background section is over one page long, while the results section is less than half a page, signalling the poor presentation of the work itself.

A list of comments is reported below.

1. The overall presentation could be (substantially) improved, including Figures quality (Fig.3(b) has low resolution, Fig.1 is unclear and incomplete).

2. The authors should provide more details about the neural network used and the training, including the one for the RF model used (is it the same from H. Liao *et al*., *Nature Machine Intelligence* 6, 1478-1486 (2024) or trained from scratch?).

3. Are the results with Clifford circuits only, or with non-Clifford as well?

4. Figure 2: why only 5 qubits? Which results are those? They seem to be the ones on random circuits similarly to H. Liao *et al*., *Nature Machine Intelligence* 6, 1478-1486 (2024), but this is not mentioned.

5. Figure 3: does it mean Global and Light cone are unnecessary? Please elaborate this part.

6. Figure 4: which values of h and J have been used? In figure 6 of H. Liao *et al*., *Nature Machine Intelligence* 6, 1478-1486 (2024), multiple configurations are considered.

7. What about extrapolation to higher Trotter steps number?